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Slack in Static Timing Analysis

Jens Vygen

***Abstract*—The notion of slack is central in static timing analysis and very large scale integration (VLSI) design in general. Negative slack means that a timing constraint is violated, while a positive slack of x ps is intended to mean that an extra delay of x ps (or a smaller delay by x ps in early mode) could be tolerated. However, this property does not hold with the standard static timing analysis model. The paper defines slack properly, shows how to compute it efficiently, and proves that it has the intended properties. The proposed idea is based on enhanced slew propagation.**

***Index Terms*—Delay budgeting, physical design, slack, slack distribution, static timing analysis, VLSI design.**

Notation Index

$S(x)$	Set of signals at x.
σ, σ'	Signals.
$\mathrm{at}(x, \sigma)$	Arrival time of signal $\sigma \in S(x)$ at x.
$\mathrm{slew}(x, \sigma)$	Slew of signal $\sigma \in S(x)$ at x.
Λ	Range of possible slews.
G	Timing graph.
ϑ_e	Delay function of arc $e \in E(G)$.
λ_e	Slew function of arc $e \in E(G)$.
Ω	Set of origin labels.
$\mathcal{S}$	Set of all signal types. $\mathcal{S} = \Omega \times \{\mathrm{early}, \mathrm{late}\} \times \{\mathrm{rise}, \mathrm{fall}\}$.
σ_e	Maps a signal type to another one when propagated along arc $e \in E(G)$.
$\delta^-(q), \delta^+(q)$	Set of arcs entering and leaving q, respectively.
ν, ν_x	Parameter in enhanced slew propagation (at x).
$\hat{\lambda}_e, \hat{\vartheta}_e$	Lipschitz constants for λ_e, ϑ_e.
T	Set of tests.
B	Balance graph.

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I. Introduction

Classical static timing analysis, though simple and efficient, has several severe drawbacks.

1) It can be overly pessimistic due to a simplified model.
2) It can also be overly optimistic: a chip might fail although static timing analysis detects no problem.
3) It works only for acyclic timing graphs. Only heuristics are known for timing graphs with directed cycles.
4) Arrival times and slacks are not continuous functions of the input. Changing an asserted arrival time slightly at some input may change a computed arrival time at some output drastically.
5) Slacks are often too optimistic. A positive slack of x ps does not mean (as one would expect) that an additional delay of x ps at this place could be tolerated without creating timing violations.

We argue that the last four mentioned drawbacks will become more and more important as extensive minimization of power consumption is done because this leads to large parts of the chip with very small slack. The first point also becomes more important and has to be addressed by statistical methods and/or transistor-level timing; however, it is not yet clear how to make this effective for large designs.

In this paper, we will show how to extend the theory of static timing analysis using enhanced slew propagation to resolve the other four problems. We propose a safe model that is as pessimistic as necessary, but not more. It probably works for general timing graphs, is continuous, and computes reliable slacks. This is a key basis for timing optimization. Moreover, computing the information is almost as simple and fast as classical propagation.

In the final part of the paper, we briefly review slack distribution (with many applications in physical design) and show how our enhanced timing model extends to optimum slack distributions in a natural and well-defined sense studied before.

This is a theoretical paper. The impact in practice will have to be determined by future work.

II. Static Timing Analysis

We briefly describe the classical static gate-level timing analysis framework essentially introduced by Hitchcock *et al.* [9], Agrawal [1], Jouppi [10], and Ousterhout [15]. We point out several weaknesses and generalize the model by enhanced slew propagation in order to remove these drawbacks. Based on the generalized model, a new theory of static timing analysis is developed.

A. Classical Propagation

In timing analysis, we check that all signals arrive at certain points within a prescribed time interval. To do this, we propagate information through the netlist.

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If all bits at registers and primary inputs remain constant from one clock cycle to the next, then the voltage remains constant everywhere on the chip. But if at least one bit changes, this information must be propagated through the network. The information that we propagate is called a signal. A signal consists of: 1) the information whether the voltage is rising (the bit changes from 0 to 1) or falling (the bit changes from 1 to 0); 2) the time when the voltage change occurs with respect to clock cycle 0; 3) a measure of how fast the voltage changes; and 4) information on the origin of the signal (a primary input or register).

The reason for 1) is the significantly different propagation delays. This partially makes up for ignoring the logical functions. As both rising and falling voltages can occur, we usually have pairs of signals. In fact, we normally have four signals at each point as we have to consider the earliest and latest possible voltage changes in each direction (rising or falling).

2) and 3) are the standard information in static timing analysis: the characteristics of a voltage change over time are encoded by two numbers: the arrival time (usually the 50% transition time) and the slew (usually the difference of 10% and 90% transition times). In the following, we will not restrict ourselves to this specific definition of arrival time and slew. However, we require that delays depend monotonely on slew. This is always the case if the point in time considered as arrival time is early enough in the voltage transition.

All arrival times of signals refer to a specific clock cycle. Since we want to check the timing conditions for the complete chip and for all cycles simultaneously, we need information 4) on the origin of the signal, in particular its local frequency.

Since the propagation delay of a signal depends on its slew (more precisely, on the voltage function over time), we have to propagate forward. So for each primary input x and each output pin x of a register, we consider a set $S(x) := \{(x, \text{early}, \text{rise}), (x, \text{early}, \text{fall}), (x, \text{late}, \text{rise}), (x, \text{late}, \text{fall})\}$ of four possible signals. With each signal $\sigma \in S(x)$ in point x, we associate its arrival time $\text{at}(x, \sigma) \in \mathbb{R}$ and some information $\text{slew}(x, \sigma) \in \Lambda$ on the shape of the voltage function. We assume that Λ (the slew range) is a finite closed interval of $\mathbb{R}$. Smaller values correspond to "stronger" signals, where the voltage goes faster from 0 to 1 (or vice versa).

Propagation is carried out in the so-called timing graph G, a directed graph whose vertex set $V(G)$ contains all pins. As usual, the elements of the arc set $E(G)$ represent connections within the gates as well as wire connections.

Each arc $e = (p, q) \in E(G)$ (also called a propagation segment) is associated with two monotonely increasing Lipschitz continuous functions $\vartheta_e : \Lambda \to \mathbb{R}$ and $\lambda_e : \Lambda \to \Lambda$. (A function $f : \Lambda \to \mathbb{R}$ is called Lipschitz continuous if there is an $\alpha \geq 0$, the Lipschitz constant, such that $|f(x) - f(y)| \leq \alpha|x - y|$ for all $x, y \in \Lambda$.) ϑ_e and λ_e describe the delay and slew at the endpoint of the propagation segment as a function of the input slew. In most cases, but not necessarily always, the value of ϑ_e will be positive; it is called the delay of the signal along e.

To allow different delays for different transitions, we use parallel arcs and introduce a mapping $\sigma_e : \mathcal{S} \to \mathcal{S} \cup \{\emptyset\}$ for each arc e, where $\mathcal{S} := \Omega \times \{\text{early}, \text{late}\} \times \{\text{rise}, \text{fall}\}$ and Ω is the set of origin labels. Usually, $\sigma_e(r, \eta, \zeta) = \emptyset$ for all $r \in \Omega$ and all $(\eta, \zeta) \neq (\eta', \zeta')$, for some $\eta' \in \{\text{early}, \text{late}\}$, $\zeta' \in \{\text{rise}, \text{fall}\}$ (meaning that the arc is used only for (η', ζ') signals), and $\sigma_e(r, \eta', \zeta') = (r, \eta', \zeta'')$ for some $\zeta'' \in \{\text{rise}, \text{fall}\}$ and all $r \in \Omega$. For each net with source p and each sink q of this net, we then have four parallel arcs (p, q), one for each element of $\{\text{early}, \text{late}\} \times \{\text{rise}, \text{fall}\}$. Moreover, for each gate, we have an internal timing subgraph defined in the timing rule.

For a start, we omit internal arcs of registers; then, the timing graph G is acyclic. So we can propagate in topological order. Let

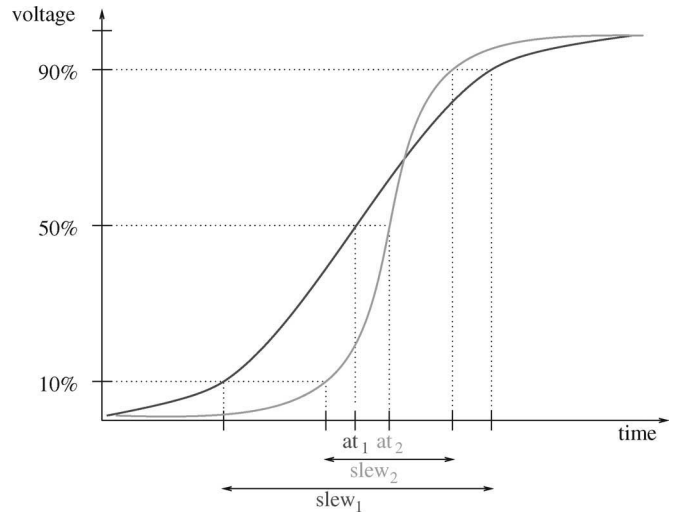

Fig. 1. Arrival times and slews.

$\delta^-(v)$ and $\delta^+(v)$ denote the sets of arcs entering and leaving vertex v, respectively. Let $q \in V(G)$ be a vertex with nonzero in-degree whose predecessors, if any, have all been processed already [i.e., we have defined a set $S(p)$ of signals for each p with $(p, q) \in \delta^-(q)$]. Then, we define a set of signals $S(e)$ for each $e = (p, q) \in \delta^-(q)$ as follows. For each signal $\sigma \in S(p)$, we add $\sigma' := \sigma_e(\sigma)$ to $S(e)$ and set

$$\begin{aligned} \text{at}(e, \sigma') &:= \text{at}(p, \sigma) + \vartheta_e\left(\text{slew}(p, \sigma)\right) \\ \text{slew}(e, \sigma') &:= \lambda_e\left(\text{slew}(p, \sigma)\right). \end{aligned} \tag{1}$$

In full path propagation, we set $S(q)$ to be the disjoint union of all $S(e)$, $e \in \delta^-(q)$, and $\text{at}(q, \sigma') := \text{at}(e, \sigma')$ and $\text{slew}(q, \sigma') := \text{slew}(e, \sigma')$. This recursively defines multisets of signals, each associated with arrival time and slew, at all vertices.

However, eventually we are interested only in the earliest or latest signal from each origin. If a signal is dominated by another one of the same type (i.e., earlier arrival time and better slew for late signals, or later arrival time and worse slew for early signals), we can safely exclude this signal from further propagation. But, if we have two signals from the same origin, say on two arcs entering the same vertex, and one of the signals has an earlier arrival time but a higher slew value than the other, it is not clear which of the two signals will arrive earlier at some remote vertex (endpoint); indeed, this can depend on the endpoint considered. Fig. 1 shows such an example of two rising signals, and their arrival time and slew according to the standard interpretations.

Since full path propagation (even when removing dominated signals) can lead to a huge number of signals, a simpler (but less accurate) way of propagation is used in practice. For each vertex p, each origin r, each $\eta \in \{\text{early}, \text{late}\}$, and each $\zeta \in \{\text{rise}, \text{fall}\}$, we consider at most one signal $(r, \eta, \zeta) \in S(p)$. Several signals differing only in arrival time or slew are united to one as follows.

We set $S(q) := \bigcup_{e \in \delta^-(q)} S(e)$, and for $\sigma \in S(q)$

$$\begin{aligned} \text{at}(q, \sigma) &:= \max\left\{\text{at}(e, \sigma) | e \in \delta^-(q), \sigma \in S(e)\right\} \\ \text{slew}(q, \sigma) &:= \max\left\{\text{slew}(e) | e \in \delta^-(q), \sigma \in S(e), \right. \\ &\qquad \left. \text{at}(e, \sigma) = \text{at}(q, \sigma)\right\} \end{aligned} \tag{2}$$

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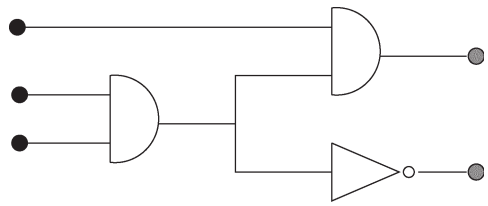

Fig. 2. Small netlist.

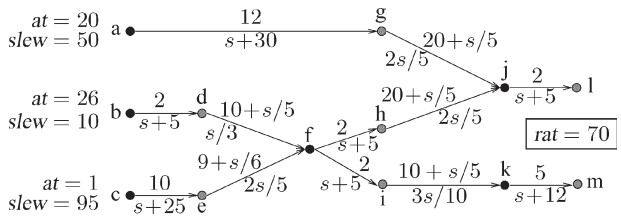

Fig. 3. Timing graph with delay and slew functions.

if σ is a late signal and

$$\begin{aligned}\mathrm{at}(q,\sigma) &:= \min\left\{\mathrm{at}(e,\sigma) | e \in \delta^-(q), \sigma \in S(e)\right\}\\ \mathrm{slew}(q,\sigma) &:= \min\left\{\mathrm{slew}(e) | e \in \delta^-(q), \sigma \in S(e),\right.\\ &\qquad \left.\mathrm{at}(e,\sigma) = \mathrm{at}(q,\sigma)\right\}\end{aligned} \tag{3}$$

if σ is an early signal.

However, ignoring all signals except the latest (earliest) one is not feasible. To illustrate this, Fig. 2 shows a simple network with three primary inputs (black), two primary outputs (grey), two ANDs, and one inverter. Fig. 3 shows the corresponding timing graph (we show only one out of four parallel arcs).

Source nodes are represented by black circles, sink nodes by grey circles. The mappings ϑ_e and λ_e are indicated above and below each arc e, respectively, as functions of $s \in \Lambda$. Moreover, we show arrival times and slews at the three primary inputs a, b, c. The required arrival time is 70 at both primary outputs l and m. Classical static timing analysis would say that this chip works, the arrival time at the critical output l is 70, just as required. But the path from c to l takes longer—a signal propagated along this path arrives at the output at time 75. Note that although delay and slew functions are nonlinear in practice, the situation is quite realistic and appears similarly many times on real chips whenever two signals arrive almost simultaneously at some point and the slightly earlier one has worse slew.

Classical static timing analysis may thus conclude that timing constraints are met although the chip will not work. Moreover, it has further disadvantages. It works only in acyclic timing graphs, arrival times are not continuous functions of design changes, and slacks are often too optimistic. Indeed, the output l in Fig. 2 has slack zero and the input c has a positive slack of 4. But if the signal at c arrived 2 units later, we would see a negative slack of -6.6 at the output l! This is in contrast to the intuitive meaning that the slack is a "safe margin": an additional delay up to this amount should not harm. See also Table I and Section III-B, where we shall return to this example and explain it in more detail.

In the following, we show how to enhance the static timing model such that all these drawbacks are resolved.

TABLE I
OLD AND NEW SLACKS

	$\nu = \infty$ (classical)				$\nu = 3$ (new)			
pin	*at*	*slew*	*rat*	slack	*at*	*slew*	*rat*	slack
a	20	50	20	0	20	50	16	-4
b	26	10	29	3	26	10	21	-5
c	1	95	5	4	1	95	-4	-5
d	28	15	31	3	28	15	23	-5
e	11	120	15	4	11	120	6	-5
f	41	5	44	3	41	45	36	-5
g	32	80	32	0	32	80	28	-4
h	43	10	46	3	43	50	38	-5
i	43	10	53	10	43	50	45	2
j	68	32	68	0	73	20	68	-5
k	55	3	65	10	63	15	65	2
l	70	37	70	0	75	25	70	-5
m	60	15	70	10	68	27	70	2

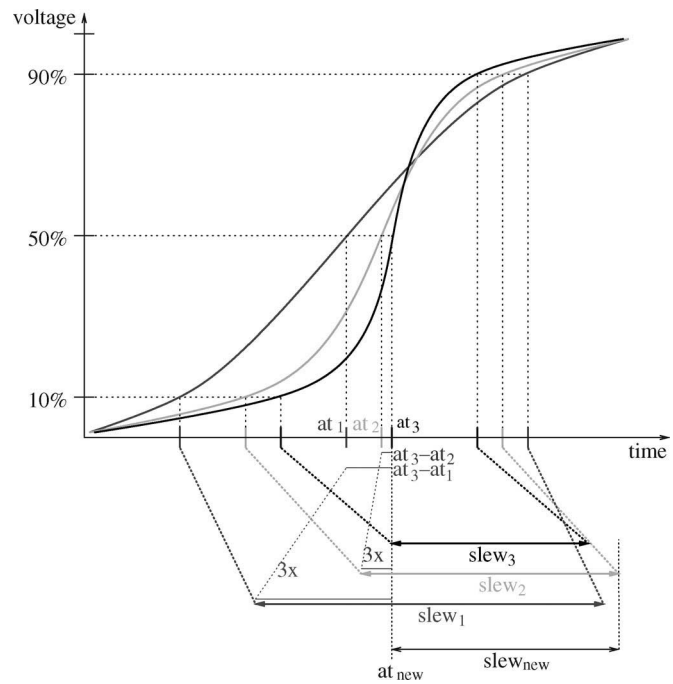

Fig. 4. Enhanced slew propagation.

B. Enhanced Slew Propagation

The following new model was proposed by Lee *et al.* [13] and Vygen [17] (independently). We set $S(q) := \bigcup_{e \in \delta^-(q)} S(e)$, and for $\sigma \in S(q)$

$$\begin{aligned}\mathrm{at}(q,\sigma) &:= \max\left\{\mathrm{at}(e,\sigma) | e \in \delta^-(q), \sigma \in S(e)\right\}\\ \mathrm{slew}(q,\sigma) &:= \max\left\{\mathrm{slew}(e,\sigma) + \nu\left(\mathrm{at}(e,\sigma) - \mathrm{at}(q,\sigma)\right) \,|\right.\\ &\qquad \left. e \in \delta^-(q), \sigma \in S(e)\right\}\end{aligned} \tag{4}$$

if σ is a late signal and

$$\begin{aligned}\mathrm{at}(q,\sigma) &:= \min\left\{\mathrm{at}(e,\sigma) | e \in \delta^-(q), \sigma \in S(e)\right\}\\ \mathrm{slew}(q,\sigma) &:= \min\left\{\mathrm{slew}(e,\sigma) + \nu\left(\mathrm{at}(e,\sigma) - \mathrm{at}(q,\sigma)\right) \,|\right.\\ &\qquad \left. e \in \delta^-(q), \sigma \in S(e)\right\}\end{aligned} \tag{5}$$

if σ is an early signal.

Here, ν is some nonnegative parameter. Fig. 4 illustrates the above formulas for late signals and $\nu = 3$. Here, the three signals are replaced by one, whose arrival time is that of the blue signal (at_3) and whose

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slew is the slew of the green signal minus three times the difference of arrival times of the blue and green signal: $\text{slew}_2 - 3(\text{at}_3 - \text{at}_2)$.

Traditional static timing analysis is equivalent to the case $\nu = \infty$ (interpreting $\infty \cdot 0 = 0$). However, as shown above, this can be too optimistic.

A safe (but usually much too pessimistic) choice is $\nu = 0$; this is sometimes called worst slew propagation and is sometimes used in practice. The following result, which has been obtained independently by Lee *et al.* [13] and Vygen [17], proposes another safe (and less pessimistic) choice.

Theorem 2.1: Let $\hat{\lambda}$ and $\hat{\vartheta}$ be global Lipschitz constants of λ_e and ϑ_e over all $e \in E(G)$ and over the whole range of Λ, i.e., $|\lambda_e(s_1) - \lambda_e(s_2)| \leq \hat{\lambda}|s_1 - s_2|$ and $|\vartheta_e(s_1) - \vartheta_e(s_2)| \leq \hat{\vartheta}|s_1 - s_2|$ for all $e \in E(G)$ and all $s_1, s_2 \in \Lambda$. Suppose that $\hat{\lambda} < 1$.

Then, using $\nu = (1 - \hat{\lambda})/\hat{\vartheta}$ in (4) and (5) leads to values $\text{at}(p, \sigma)$ for each p and each late signal (early signal) $\sigma \in S(p)$ that are at least (at most) those of any signal σ propagated to p in full path propagation.

If G is bipartite, i.e., the source and sink pins alternate on every path, we can strengthen the above result (see Theorem 2.3).

Since it is desirable to choose ν as large as possible (while remaining pessimistic in comparison with full-path propagation), it might be better to have an individual number ν_p for each vertex p; these numbers should satisfy $(1/\nu_p) \geq \hat{\vartheta}_e + (\hat{\lambda}_e/\nu_q)$ for every arc $e = (p, q)$ (see Theorem 2.2). For acyclic timing graphs, such values can be computed easily by backward propagation. It is even questionable whether the maximum has to be taken over the whole range of Λ.

Note that the above result can also be used to improve full path propagation. Given two late signals σ and σ' of the same type in some vertex q, we can eliminate σ (it is dominated by σ') if $\text{at}(\sigma) + (1/\nu_q)\max\{0, \text{slew}(\sigma) - \text{slew}(\sigma')\} \leq \text{at}(\sigma')$. This may reduce the number of signals to be propagated significantly. For example, in Fig. 4 (for $\nu = 3$), the green signal (number 2) dominates the red one (number 1). We shall discuss this again in Section II-D.

Besides the fact that classical static timing analysis ($\nu = \infty$) is too optimistic, our model has more advantages. We can handle nonacyclic graphs in a satisfactory way (cf. Theorem 2.5), and we can determine the critical parts of a chip more accurately (cf. Theorems 3.1 and 3.2). Indeed, the more paths are critical (e.g., after minimizing power consumption), the more mistaken is the traditional model. Moreover, we have a continuous behavior (small changes in the input lead to small changes of arrival times), a fact that reflects real behavior and will prove useful for optimization.

Note that for ramp functions, $\nu = 2$ in (4) and (5) corresponds to the latest (respectively, earliest) 90% transition time (for falling signals: 10% transition time). This is essentially the pruning scheme of Blaauw *et al.* [4]. They argued that ν can be chosen to be at least 2 (see also [13]). Of course, Theorem 2.1 (and its strengthening, Theorem 2.3) allows an even better choice.

C. Required Arrival Times and Tests

After having computed arrival times (and slew), we can compare them with required arrival times at primary outputs and input pins of registers. These are the vertices of the timing graph with zero out-degree. So for each pair (x, y) such that y (a primary output or an input pin of a register) receives a signal from x (a primary input or an output pin of a register), and each $\sigma \in S(x)$, we have to check

$$\begin{aligned} \text{at}\,(y, (x, \text{late}, \zeta)) &\leq \text{rat}(y, \text{late}, \zeta) + \text{adjust}(x, y, \text{late}) \\ \text{at}\,(y, (x, \text{early}, \zeta)) &\geq \text{rat}(y, \text{early}, \zeta) + \text{adjust}(x, y, \text{early}). \end{aligned} \quad (6)$$

We do not go into detail how the adjusts are computed. In the simplest case, when x and y belong to the same clock domain, $\text{adjust}(x, y, \text{late})$ is one clock cycle and $\text{adjust}(x, y, \text{early})$ is zero. Different frequencies and more complicated timing models, transparent latches, false paths, and multicycle paths lead to different adjusts. An important implementation detail is that we do not need two different origin labels for the signals starting in x_1 and x_2 if $\text{adjust}(x_1, y, \eta) = \text{adjust}(x_2, y, \eta)$ for $\eta =$ early, late and all y that are reachable from both x_1 and x_2. This leads to a substantial reduction in memory and runtime consumption.

In a more general timing model, arrival times and required arrival times at registers are not fixed in advance. The timing graph is then a general directed graph, not necessarily acyclic. We still have arrival times and slews for signals $S(x)$ for all vertices x with zero in-degree and required arrival times (and slews) at vertices of zero out-degree. For other vertices, we may or may not have required arrival times. Other timing constraints can be modeled using so-called tests: a test is an ordered pair $e = ((x, \sigma), (y, \sigma'))$ that represents the constraint that $\text{at}(x, \sigma) \leq \text{at}(y, \sigma')$. For example, each register has at least two tests, a setup test and a hold test. After having computed all arrival times, the inequalities defined by tests can be checked easily.

In the following, we ignore adjusts for the sake of a simpler notation. Note, however, that they can be included easily without any complication.

General timing graphs with test arcs allow us to encode that an arbitrary signal must arrive before another one. Thus, one can include clock signals in timing analysis without having to fix their arrival times *a priori*. Moreover, circuits with complex internal structures, dynamic logic, and asynchronous signals can be handled by introducing appropriate tests.

D. General Static Timing Analysis

The above considerations lead to the following formulation, which is general enough to capture all situations arising in common industrial design practice.

General Static Timing Analysis Problem

INSTANCE:

1) A finite set Ω (of origin labels). Define $\mathcal{S} := \Omega \times \{\text{early, late}\} \times \{\text{rise, fall}\}$.
2) A directed graph G, whose arcs are called propagation segments.
3) A binary relation $T \subseteq (V(G) \times \mathcal{S})^2$, whose elements are called tests.
4) A compact interval $\Lambda \subseteq \mathbb{R}$. For each $e \in E(G)$, two Lipschitz continuous and monotonely increasing functions $\vartheta_e : \Lambda \to \mathbb{R}$ and $\lambda_e : \Lambda \to \Lambda$.
5) For each $e \in E(G)$, a mapping $\sigma_e : \mathcal{S} \to \mathcal{S} \cup \{\emptyset\}$, such that $\sigma_e(r, \eta, \zeta) \in (\Omega \times \{\eta\} \times \{\text{rise, fall}\}) \cup \{\emptyset\}$ for all $(r, \eta, \zeta) \in \mathcal{S}$.
6) For each vertex $x \in V(G)$ with zero in-degree: a set $S(x)$ of elements of $\mathcal{S}$ and numbers $\text{at}(x, \sigma) \in \mathbb{R}$, $\text{slew}(x, \sigma) \in \Lambda$ for each $\sigma \in S(x)$.
7) For each vertex $x \in V(G)$ with zero out-degree numbers $\text{rat}(x, \sigma) \in \mathbb{R}$ for all $\sigma \in \mathcal{S}$.
8) A number $\nu_x \in \mathbb{R}_+ \cup \{\infty\}$ for each $x \in V(G)$.

TASK:

Find sets $S(x) \subseteq \mathcal{S}$ and numbers $\text{at}(x, \sigma) \in \mathbb{R}$ and $\text{slew}(x, \sigma) \in \Lambda$ ($\sigma \in S(x)$) for each vertex $x \in V(G)$ with nonzero in-degree, such that the following conditions hold, or decide that they do not exist.

1) For each $e = (p, q) \in E(G)$ and $\sigma \in S(p)$ with $\sigma' := \sigma_e(\sigma) \in \mathcal{S}$, we have $\sigma' \in S(q)$ and

$$\frac{1}{\nu_q} \max\{0, \lambda_e(\text{slew}(p, \sigma)) - \text{slew}(q, \sigma')\} + \text{at}(p, \sigma) + \vartheta_e(\text{slew}(p, \sigma)) \leq \text{at}(q, \sigma') \quad (7)$$

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if σ is a late signal and

$$\frac{1}{\nu_q}\min\{0, \lambda_e(\text{slew}(p,\sigma)) - \text{slew}(q,\sigma')\} + \text{at}(p,\sigma) + \vartheta_e(\text{slew}(p,\sigma)) \geq \text{at}(q,\sigma') \quad (8)$$

if σ is an early signal.

2) For each $t = ((p,\sigma),(q,\sigma')) \in T$ with $\sigma \in S(p)$ and $\sigma' \in S(q)$ we have

$$\text{at}(p,\sigma) \leq \text{at}(q,\sigma'). \quad (9)$$

3) For each vertex x with zero out-degree and $\sigma \in S(x)$ we have

$$\text{at}(x,\sigma) \leq \text{rat}(x,\sigma) \quad (10)$$

if σ is a late signal and

$$\text{at}(x,\sigma) \geq \text{rat}(x,\sigma) \quad (11)$$

if σ is an early signal.

If $\nu_q = 0$ or $\nu_q = \infty$, (7) and (8)—and similar inequalities in the following—are to be interpreted in the sense that they must hold for every positive ν_q or for sufficiently large ν_q, respectively.

In practice, some delays are often associated with the tests (e.g., setup and hold time). This gives an additional number on one side of each inequality (9); this number may depend on the slews of both signals involved. However, this situation can easily be transformed into the above model by introducing auxiliary propagation segments adjacent to each test arc.

We did not include required slews at vertices of zero out-degree in our model. These could either be included by strengthening conditions (10) and (11), or again, without changing the model, by auxiliary propagation segments. Moreover, noninverting loop-back paths and common clocktree path pessimism removal can be dealt with within the presented model with the usual techniques.

Definition 2.1: Let $\sigma = (r,\eta,\zeta)$ and $\sigma' = (r',\eta',\zeta')$ be two signals at some vertex x. We say that σ' *dominates* σ if $r = r'$, $\eta = \eta'$, $\zeta = \zeta'$, and the following holds:

$$\text{at}(x,\sigma) + \frac{1}{\nu_x}\max\{0, \text{slew}(x,\sigma) - \text{slew}(x,\sigma')\} \leq \text{at}(x,\sigma') \quad (12)$$

and $\eta =$ late, or

$$\text{at}(x,\sigma) + \frac{1}{\nu_x}\min\{0, \text{slew}(x,\sigma) - \text{slew}(x,\sigma')\} \geq \text{at}(x,\sigma') \quad (13)$$

and $\eta =$ early.

The justification of this domination rule, which evidently defines a partial order on $\mathbb{R} \times \Lambda$, lies in the safe choice of ν_x.

Definition 2.2: Given an instance of the general static timing analysis problem, let $\tilde{S}(x)$, $x \in V(G)$, be the minimal signal sets that can be part of a feasible solution.

We define the propagation graph as the graph with vertices (x,σ), $x \in V(G)$, $\sigma \in \tilde{S}(x)$, and an arc $((p,\sigma),(q,\sigma'))$ for each $(p,q) \in E(G)$, $\sigma \in \tilde{S}(p)$, and $\sigma' = \sigma_e(\sigma) \in \tilde{S}(q)$. The induced subgraph on the vertices corresponding to η signals is called the η propagation graph ($\eta =$ early, late).

For $\eta \in \{\text{early, late}\}$, an η propagation sequence is a path in the η propagation graph. If the first and last vertices coincide, the propagation sequence is called cyclic.

The numbers ν_x and $x \in V(G)$ are safe if for every propagation sequence $((x_1,\sigma_1), e_1, (x_2,\sigma_2), e_2, \ldots, e_{k-1}, (x_k,\sigma_k))$ (late or early) and for any number $a_{i,j}$, $s_{i,j}$ $(i = 1,2, j = 1,\ldots,k)$ with $a_{i,j+1} = a_{i,j} + \vartheta_{e_j}(s_{i,j})$, $s_{i,j+1} = \lambda_{e_j}(s_{i,j})(i = 1,2, j = 1,\ldots,k-1)$, and $a_{1,1} \geq a_{2,1} + (1/\nu_{x_1})\max\{0, s_{2,1} - s_{1,1}\}$ we have $a_{1,k} \geq a_{2,k}$.

In particular, choosing $\nu_x = 0$ for all $x \in V(G)$ is always safe due to the monotonicity of the functions ϑ_e and λ_e $(e \in E(G))$. As long as the numbers ν_x $(x \in V(G))$ are safe, elimination of dominated signals cannot hurt. Of course we want the numbers ν_x $(x \in V(G))$ to be as large as possible as long as they are safe. A possible choice was proposed by Theorem 2.1. This result can be generalized as follows.

Theorem 2.2: Let an instance of the general static timing analysis problem be given. For $e \in E(G)$, let $\hat{\lambda}_e$ and $\hat{\vartheta}_e$ be Lipschitz constants, i.e., $|\lambda_e(s_1) - \lambda_e(s_2)| \leq \hat{\lambda}_e|s_1 - s_2|$ and $|\vartheta_e(s_1) - \vartheta_e(s_2)| \leq \hat{\vartheta}_e|s_1 - s_2|$ for all s_1, $s_2 \in \Lambda$. Suppose that for all $e = (x,y) \in E(G)$

$$\frac{1}{\nu_x} \geq \hat{\vartheta}_e + \frac{\hat{\lambda}_e}{\nu_y}. \quad (14)$$

Then, the numbers ν_x $x \in V(G)$, are safe.

Proof: Let $e = (x,y) \in E(G)$, and take any numbers a_1, $a_2 \in \mathbb{R}$, s_1, $s_2 \in \Lambda$ with $a_1 \geq a_2 + (1/\nu_x)\max\{0, s_2 - s_1\}$. Then

$$\begin{aligned} a_1 + \vartheta_e(s_1) &\geq a_2 + \frac{1}{\nu_x}\max\{0, s_2 - s_1\} + \vartheta_e(s_1) \\ &\geq a_2 + \frac{1}{\nu_x}\max\{0, s_2 - s_1\} + \vartheta_e(s_2) \\ &\quad - \hat{\vartheta}_e\max\{0, s_2 - s_1\} \\ &\geq a_2 + \vartheta_e(s_2) + \frac{\hat{\lambda}_e}{\nu_y}\max\{0, s_2 - s_1\} \\ &\geq a_2 + \vartheta_e(s_2) + \frac{1}{\nu_y}\max\{0, \lambda_e(s_2) - \lambda_e(s_1)\}. \end{aligned} \quad (15)$$

This directly implies safeness. ■

This result (in a similar form) can be found in [13] and [17], and implicitly in [4]. We call numbers ν_x, $x \in V(G)$, strongly safe if they satisfy (14). Now the following strengthening of Theorem 2.1 follows easily.

Theorem 2.3: Given an instance of the general static timing analysis problem and partitions $V(G) = V_1 \dot\cup V_2$ and $E(G) = E_1 \dot\cup E_2$ such that E_i is the set of arcs with tail in V_{3-i} and head in V_i $(i = 1, 2)$. Using the notation of Theorem 2.2, let $\hat{\lambda}_{(i)} \geq \max\{\hat{\lambda}_e | e \in E_i\}$ and $\hat{\vartheta}_{(i)} \geq \max\{\hat{\vartheta}_e | e \in E_i\}$ be global Lipschitz constants. Then, the numbers

$$\nu_x = \max\left\{0, \frac{1 - \hat{\lambda}_{(1)}\hat{\lambda}_{(2)}}{\hat{\vartheta}_{(3-i)} + \hat{\lambda}_{(3-i)}\hat{\vartheta}_{(i)}}\right\} \quad (16)$$

for $x \in V_i$ $(i = 1,2)$ are strongly safe.

Proof: For $e = (x,y) \in E_i$, we have $(1/\nu_x) = \hat{\vartheta}_{(i)} + (\hat{\lambda}_{(i)}/\nu_y)$, implying (14). Now apply Theorem 2.2. ■

Note that with Elmore delay, we have $\hat{\lambda}_{(1)} = 1$ and $\hat{\vartheta}_{(1)} = 0$ (where E_1 are the net arcs); thus, $\nu = (1 - \hat{\lambda}_{(2)})/\hat{\vartheta}_{(2)}$. An analysis of a recent CMOS library [13] suggested that $\hat{\lambda}_{(2)} \approx 0.37$ and $\hat{\vartheta}_{(2)} \approx 0.22$, yielding $\nu \approx 2.86$.

It would cause no difficulty to introduce two numbers ν_x for each vertex x, one for early signals and one for late signals. However, this would not gain much as the optimal Lipschitz constants for late and early arcs are similar for practical instances. Therefore, we prefer the simpler notation.

We will apply iterative propagation. But let us first give a simpler problem formulation, without required arrival times and tests. We need the following notation.

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Definition 2.3: Let $S(x)$, $S'(x)$ be sets of elements of $\mathcal{S}$, and let $\mathrm{at}(x,\sigma) \in \mathbb{R}$, $\mathrm{slew}(x,\sigma) \in \Lambda(\sigma \in S(x))$, $\mathrm{at}(x,\sigma') \in \mathbb{R}$, and $\mathrm{slew}(x,\sigma') \in \Lambda(\sigma' \in S'(x))$ be given for each vertex $x \in V(G)$.

We say that $S(x)$, $\mathrm{at}(x,\sigma)$, $\mathrm{slew}(x,\sigma)$ $(x \in V(G), \sigma \in S(x))$ is dominated by $S'(x)$, $\mathrm{at}(x,\sigma')$, $\mathrm{slew}(x,\sigma')$ $(x \in V(G), \sigma' \in S'(x))$ if for each $x \in V(G)$ and each $\sigma \in S(x)$ we have a $\sigma' \in S'(x)$ dominating σ.

Evidently, this defines a partial order. We look for solutions that are minimal with respect to this partial order.

Arrival Time Problem

INSTANCE:

An instance of the general static timing analysis problem, where the numbers ν_x, $x \in V(G)$ are strongly safe.

TASK:

Find sets $S(x) \subseteq \mathcal{S}$ and numbers $\mathrm{at}(x,\sigma) \in \mathbb{R}$ and $\mathrm{slew}(x,\sigma) \in \Lambda(\sigma \in S(x))$ for each vertex $x \in V(G)$ with nonzero in-degree such that the following conditions hold (or decide that none exists).

For each $e = (p,q) \in E(G)$ and $\sigma \in S(p)$ with $\sigma' = \sigma_e(\sigma) \in \mathcal{S}$, we have $\sigma' \in S(q)$ and

$$\frac{1}{\nu_q} \max\{0, \lambda_e(\mathrm{slew}(p,\sigma)) - \mathrm{slew}(q,\sigma')\} + \mathrm{at}(p,\sigma) + \vartheta_e(\mathrm{slew}(p,\sigma)) \leq \mathrm{at}(q,\sigma') \quad (17)$$

if σ is a late signal and

$$\frac{1}{\nu_q} \min\{0, \lambda_e(\mathrm{slew}(p,\sigma)) - \mathrm{slew}(q,\sigma')\} + \mathrm{at}(p,\sigma) + \vartheta_e(\mathrm{slew}(p,\sigma)) \geq \mathrm{at}(q,\sigma') \quad (18)$$

if σ is an early signal.

Moreover, the solution must be dominated by any other such solution.

We shall prove that if there is a solution satisfying (17) and (18), then there is also one that is dominated by any other such solution. Moreover, we shall show how to solve this problem.

It is clear that solving the arrival time problem is sufficient. Either such a solution meets all further constraints (tests and required arrival times) specified in the instance of the general static timing analysis problem, or the instance is infeasible.

E. Solving the Arrival Time Problem

In this section, we discuss how to solve the arrival time problem. Let us start with the acyclic case. The following notation will prove useful. For $q \in V(G)$ and $\sigma' \in \tilde{S}(q)$, let $\Xi(q,\sigma')$ be the set of pairs $(e,\sigma) \in \delta_G^-(q) \times \mathcal{S}$ with $e = (p,q)$, $\sigma \in S(p)$, and $\sigma_e(\sigma) = \sigma'$.

Theorem 2.4: The arrival time problem can be solved in $O(|\Omega||E(G)|)$ time if the propagation graph is acyclic.

Proof: We proceed as indicated in (1), (4), and (5). For each pair $(e,\sigma) \in \Xi(q,\sigma')$, we set $\mathrm{at}(e,\sigma) := \mathrm{at}(p,\sigma) + \vartheta_e(\mathrm{slew}(p,\sigma))$ and $\mathrm{slew}(e,\sigma) := \lambda_e(\mathrm{slew}(p,\sigma))$. Then we add only one copy of σ' to $S(q)$ and set

$$\begin{aligned}\mathrm{at}(q,\sigma') &:= \max\{\mathrm{at}(e,\sigma) | (e,\sigma) \in \Xi(q,\sigma')\}\\ \mathrm{slew}(q,\sigma') &:= \max\{\mathrm{slew}(e,\sigma) + \nu_q(\mathrm{at}(e,\sigma) - \mathrm{at}(q,\sigma')) \mid \\ &\qquad (e,\sigma) \in \Xi(q,\sigma')\}\end{aligned} \quad (19)$$

if σ' is a late signal, and

$$\begin{aligned}\mathrm{at}(q,\sigma') &:= \min\{\mathrm{at}(e,\sigma) | (e,\sigma) \in \Xi(q,\sigma')\}\\ \mathrm{slew}(q,\sigma') &:= \min\{\mathrm{slew}(e,\sigma) + \nu_q(\mathrm{at}(e,\sigma) - \mathrm{at}(q,\sigma')) \mid \\ &\qquad (e,\sigma) \in \Xi(q,\sigma')\}\end{aligned} \quad (20)$$

if σ' is an early signal. It is clear that this satisfies constraints (17) and (18) for arcs entering q.

The claim that none of $\mathrm{at}(q,\sigma')$ and $\mathrm{at}(q,\sigma') + (1/\nu_q)\mathrm{slew}(q,\sigma')$ can be chosen smaller (greater) for any late signal (early signal) $\sigma' \in S(q)$ follows easily by induction (again in topological order of the propagation graph). The running time is easy to check. ■

The situation is more difficult for nonacyclic graphs. We proceed in the most natural way and iterate the propagation.

Iterative Propagation

INPUT: An instance of the general static timing analysis problem.

OUTPUT: A feasible solution or the decision that none exists.

1) Choose a minimal set $X \subseteq V(G)$ such that $G - X$ is acyclic.
2) Set $S(x) := \emptyset$ for $x \in X$.
3) Propagate in $G - \delta^-(X)$, i.e., solve the resulting instance of the arrival time problem according to Theorem 2.4.
4) Propagate on $\delta^-(X)$ and update $S(x)$ and $\mathrm{at}(x,\sigma)$, $\mathrm{slew}(x,\sigma)$ $(\sigma \in S(x))$ for $x \in X$. If at least one of these values changes, go to 3).

This gives a sequence of vectors of sets of signals. For general values of ν_x (in particular in the classical case $\nu_x = \infty$ for all x), this sequence has no monotonicity properties, and it is not clear whether it converges. However, if the numbers ν_x, $x \in V(G)$, are strongly safe, we have some monotonicity, in the sense of Definition 2.3.

Theorem 2.5: Let I_1 and I_2 be two instances of the arrival time problem on an acyclic graph, differing only in the information on the vertices with zero in-degree. Suppose that I_1 dominates I_2 with respect to the information $S(x)$, $\mathrm{at}(x,\sigma)$, $\mathrm{slew}(x,\sigma)$ $(\sigma \in S(x))$ for vertices x of zero in-degree. Then the solution to I_1 dominates the solution to I_2.

Proof: Straightforward induction using (15). ■

Consider the sequence of vectors with entries $\mathrm{at}(x,\sigma)$ indexed by the pairs (x,σ) for $x \in X$ and $\sigma \in \tilde{S}(x)$. By Theorem 2.5, this sequence is monotone. Hence, it is either unbounded (in which case the instance of the general static timing analysis problem is infeasible), or it converges. (It is a well-known theorem that every bounded monotone sequence converges.) If the above sequence converges, the sequence of the slews is convergent too. We summarize:

Theorem 2.6: A feasible instance of the arrival time problem has a unique solution. The sequence of signals, arrival times, and slews computed by iterative propagation converges to this solution. For an infeasible instance of the arrival time problem, it is even impossible to satisfy only (17) and (18).

Either this solution constitutes a feasible solution to the general static timing analysis problem, or there is no feasible solution. It is easy to construct (artificial) examples where no feasible solution is found after any finite number of iterations, although one exists. However, this is very unlikely in practice. The author's experience with iterative propagation in classical timing analysis on industrial chips suggests that solutions of excellent accuracy can be obtained after a very small number of iterations.

III. SLACK

In the design process, it is often useful to know which are the critical parts of a chip. In particular, we ask how much additional delay we

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could spend at a certain point without violating any constraint; in other words, how much slack a constraint has. We shall now make this notion precise and prove important properties.

A. New Definition

Assume an instance of the general static timing analysis problem. For a vertex x of the timing graph, we define the early slack and late slack of x by

$$\begin{aligned}\text{slack}(x,\text{early}) &:= \min\{\text{at}(x,\sigma)-\text{rat}(x,\sigma)|\sigma\in S(x)\text{ early signal}\}\\ \text{slack}(x,\text{late}) &:= \min\{\text{rat}(x,\sigma)-\text{at}(x,\sigma)|\sigma\in S(x)\text{ late signal}\}.\end{aligned} \tag{21}$$

We assume that the arrival times are chosen minimal (maximal) for all late (early) signals, as this implies maximal slacks. In other words, we assume a solution to the arrival time problem. But definition, (21) also involves the required arrival times; these are computed by a kind of backward propagation using the following recursion formulas.

For a propagation segment $e=(p,q)$, $\sigma\in S(p)$ and $\sigma'=\sigma_e(\sigma)\in S(q)$, we set

$$\begin{aligned}\text{rat}(e,\sigma) := \text{rat}(q,\sigma') &- \vartheta_e(\text{slew}(p,\sigma))\\ &- \frac{1}{\nu_q}\max\{0,\lambda_e(\text{slew}(p,\sigma))-\text{slew}(q,\sigma')\}\end{aligned} \tag{22}$$

if σ is a late signal, and

$$\begin{aligned}\text{rat}(e,\sigma) := \text{rat}(q,\sigma') &- \vartheta_e(\text{slew}(p,\sigma))\\ &- \frac{1}{\nu_q}\min\{0,\lambda_e(\text{slew}(p,\sigma))-\text{slew}(q,\sigma')\}\end{aligned} \tag{23}$$

if σ is an early signal.

For a vertex p with nonzero out-degree in G and a late signal $\sigma\in S(p)$, we set

$$\begin{aligned}\text{rat}(p,\sigma) := \min\Big\{&\min\left\{\text{rat}(e,\sigma)|e\in\delta_G^+(p),\sigma_e(\sigma)\neq\emptyset\right\},\\ &\min\{\text{at}(q,\sigma')|((p,\sigma),(q,\sigma'))\in T,\\ &\sigma'\in S(q)\}\Big\}.\end{aligned} \tag{24}$$

For a vertex p with nonzero out-degree in G and an early signal $\sigma\in S(p)$, we set

$$\begin{aligned}\text{rat}(p,\sigma) := \max\Big\{&\max\left\{\text{rat}(e,\sigma)|e\in\delta_G^+(p),\sigma_e(\sigma)\neq\emptyset\right\},\\ &\max\{\text{at}(q,\sigma')|((q,\sigma'),(p,\sigma))\in T,\\ &\sigma'\in S(q)\}\Big\}.\end{aligned} \tag{25}$$

For a vertex p with zero out-degree in G, the numbers $\text{rat}(p,\sigma)$ are given as input. If there are tests incident to such a vertex, we take the minimum of (24) and the given $\text{rat}(p,\sigma)$, or the maximum of (25) and the given $\text{rat}(p,\sigma)$, respectively. If the timing graph is acyclic, this recursively defines required arrival times (and thus slacks) everywhere, and they can be computed in linear time.

Again, for $\nu_x=\infty(x\in V(G))$, this coincides with the classical slack computation. But, as noted in Section II-A, these slacks do not have the intended property.

Fortunately, if the numbers ν_x $(x\in V(G))$ are strongly safe, then the slacks as defined above have the intended property. We can add as much delay as $\text{slack}(p,\text{late})$ in a pin p without making any path containing p too slow. Before proving this essential property, we show that our concept of slacks also works for nonacyclic timing graphs.

Indeed, the above definitions extend to general timing graphs without any change as long as a feasible solution to the arrival time problem exists. Required arrival times satisfying (22)–(25) can be computed easily by a shortest path computation. We need two prerequisites. First, a primary output or a test must be reachable from every vertex. This holds unless the vertex is trivially redundant. Second, the sum of $\text{rat}(e_i,\sigma_i)-\text{rat}(v_{i+1},\sigma_{i+1})$, $i=1,\ldots,k-1$, as defined by (22), must be nonnegative on every cyclic late propagation sequence

$$((v_1,\sigma_1),e_1,(v_2,\sigma_2),e_2,\ldots,e_{k-1},(v_k,\sigma_k)) \tag{26}$$

(and, analogously, nonpositive on every cyclic early propagation sequence). However, note that we have for $i=1,\ldots,k-1$

$$\text{rat}(e_i,\sigma_i)-\text{rat}(v_{i+1},\sigma_{i+1})\geq\text{at}(v_i,\sigma_i)-\text{at}(v_{i+1},\sigma_{i+1}) \tag{27}$$

[due to (22) and (17)], and the sum of the latter values on a cyclic propagation sequence is zero. For early propagation sequences, the same argument applies with reversed inequality sign.

With the Moore–Bellman–Ford shortest path algorithm (see, for example, [11]), we get the following.

Proposition 3.1: Required arrival times, and thus slacks, can be computed in $O(|V(G)||E(G)|+|T|)$ time provided that a feasible solution to the arrival time problem is given.

B. Example

Let us consider the example in Fig. 2 again. Table I shows arrival times, slews, required arrival times, and late slacks for the 13 pins a–m with classical propagation ($\nu=\infty$, columns 2–5) and with enhanced slew propagation and the required arrival definition of Section III-A with $\nu_x=\nu=3$ for all x (columns 6–9).

All classical slack values are too optimistic in this example. For example, increasing the arrival time at input c by 3 units (4 instead of 1) and applying classical timing analysis yields the surprising result that all slacks, except those at a and g, become suddenly negative! Such awkward results cannot occur with the new slacks as proposed here, as the model is continuous. Moreover, positive slacks guarantee a safe margin within which delay changes cause no harm. This will be proved in the next section.

Note that the notion of critical path does not make sense anymore: the subgraph induced by pins with smallest slack value is, in general, not a path.

C. Main Properties

Now we prove the essential properties of the so-defined slacks.

Lemma 3.1: Let $q\in V(G)$ be a vertex with nonzero in-degree and $\sigma'\in S(q)$. For each $(e,\sigma)\in\Xi(q,\sigma')$, we have $\text{rat}(e,\sigma)-\text{at}(p,\sigma)\geq\text{rat}(q,\sigma')-\text{at}(q,\sigma')$ if e is a late arc, and $\text{rat}(e,\sigma)-\text{at}(p,\sigma)\leq\text{rat}(q,\sigma')-\text{at}(q,\sigma')$ if e is an early arc. Moreover, there exists an $(e,\sigma)\in\Xi(q,\sigma')$ with $\text{rat}(e,\sigma)-\text{at}(p,\sigma)=\text{rat}(q,\sigma')-\text{at}(q,\sigma')$.

Proof: If e is a late arc, we have

$$\begin{aligned}\text{rat}(e,\sigma)-\text{at}(p,\sigma) &= \text{rat}(q,\sigma')-\text{at}(p,\sigma)-\vartheta_e(\text{slew}(p,\sigma))\\ &\quad-\frac{1}{\nu_q}\max\{0,\lambda_e(\text{slew}(p,\sigma))-\text{slew}(q,\sigma')\}\\ &\geq \text{rat}(q,\sigma')-\text{at}(q,\sigma').\end{aligned} \tag{28}$$

If e is an early arc, we have

$$\begin{aligned}\text{rat}(e,\sigma)-\text{at}(p,\sigma) &= \text{rat}(q,\sigma')-\text{at}(p,\sigma)-\vartheta_e(\text{slew}(p,\sigma))\\ &\quad-\frac{1}{\nu_q}\min\{0,\lambda_e(\text{slew}(p,\sigma))-\text{slew}(q,\sigma')\}\\ &\leq \text{rat}(q,\sigma')-\text{at}(q,\sigma').\end{aligned} \tag{29}$$

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We have equality for some $(e,\sigma) \in \Xi(q,\sigma')$ because of the minimum/maximum possible choice of $\mathrm{at}(q,\sigma')$ and $\mathrm{at}(q,\sigma') + (1/\nu_q)\mathrm{slew}(q,\sigma')$ in a solution to the arrival time problem. ■

Proposition 3.2: The η slack at a vertex of nonzero out-degree in G, if not determined by a test incident to this vertex, is at least the minimum η slack of its successors ($\eta \in \{\text{early}, \text{late}\}$).

Proof: Let $p \in V(G)$. Let $\sigma \in S(p)$ be a late signal with $\mathrm{slack}(p,\text{late}) = \mathrm{rat}(p,\sigma) - \mathrm{at}(p,\sigma)$, and let $e = (p,q) \in \delta_G^+(p)$ with $\mathrm{rat}(p,\sigma) = \mathrm{rat}(e,\sigma)$. Then (using Lemma 3.1)

$$\begin{aligned}\mathrm{slack}(p,\text{late}) &= \mathrm{rat}(p,\sigma) - \mathrm{at}(p,\sigma)\\ &= \mathrm{rat}(e,\sigma) - \mathrm{at}(p,\sigma)\\ &\geq \mathrm{rat}(q,\sigma') - \mathrm{at}(q,\sigma')\\ &\geq \mathrm{slack}(q,\text{late})\end{aligned} \tag{30}$$

where $\sigma' := \sigma_e(\sigma)$. Similar for early signals. ■

Proposition 3.3: The η slack at a vertex of nonzero in-degree in G is at least the minimum η slack of its predecessors ($\eta \in \{\text{early}, \text{late}\}$).

Proof: Let q be a vertex of nonzero in-degree. Let $\sigma' \in S(q)$ be a late signal with $\mathrm{slack}(q,\text{late}) = \mathrm{rat}(q,\sigma') - \mathrm{at}(q,\sigma')$. Let $(e = (p,q),\sigma) \in \Xi(q,\sigma')$ such that $\mathrm{rat}(e,\sigma) - \mathrm{at}(p,\sigma) = \mathrm{rat}(q,\sigma') - \mathrm{at}(q,\sigma')$ (cf. Lemma 3.1). Then

$$\begin{aligned}\mathrm{slack}(q,\text{late}) &= \mathrm{rat}(q,\sigma') - \mathrm{at}(q,\sigma')\\ &= \mathrm{rat}(e,\sigma) - \mathrm{at}(p,\sigma)\\ &\geq \mathrm{rat}(p,\sigma) - \mathrm{at}(p,\sigma)\\ &\geq \mathrm{slack}(p,\text{late}).\end{aligned} \tag{31}$$

Similar for early signals. ■

We now prove our main result on slacks.

Theorem 3.1: Consider a timing graph where all late slacks are nonnegative. Let X be a set of vertices such that no propagation sequence contains more than one of them (an antichain). Suppose we add a delay of $d(x) \leq \mathrm{slack}(x,\text{late})$ at each vertex $x \in X$. Then all late slacks with respect to the solution to the modified instance of the arrival time problem remain nonnegative.

Proof: The modified instance of the arrival time problem remains feasible. For instance, we obtain a solution satisfying (17) by adding $\sum_{x\in X} d(x)$ to each late arrival time at all vertices that are reachable from X along a propagation sequence; hence, Theorem 2.6 implies feasibility.

Denote by at' and slew' the new arrival times and slews (the solution to the modified instance of the arrival time problem). We claim that

$$\mathrm{at}'(p,\sigma) = \mathrm{at}(p,\sigma) \text{ and } \mathrm{slew}'(p,\sigma) = \mathrm{slew}(p,\sigma) \tag{32}$$

for all vertices p that are not reachable from X and $\sigma \in S(p)$, and

$$\mathrm{at}'(p,\sigma) \leq \mathrm{rat}(p,\sigma) - \frac{1}{\nu_p}\max\{0, \mathrm{slew}'(p,\sigma) - \mathrm{slew}(p,\sigma)\} \tag{33}$$

for all vertices p that are reachable from X and all $\sigma \in S(p)$. (Note that rat denotes the original required arrival times, i.e., with respect to the original values at and slew.)

The above inequalities imply $\mathrm{at}'(p,\sigma) \leq \mathrm{rat}(p,\sigma)$ everywhere, in particular at vertices with zero out-degree and wherever $\mathrm{rat}(p,\sigma)$ is determined by a test. Since these required arrival times remain unchanged, Proposition 3.2 then implies the theorem.

We prove (32) and (33) by induction on the steps of iterative propagation. We show that the claim holds at any stage. Since the arrival times in iterative propagation converge to the unique solution of the modified instance of the arrival time problem by Theorem 2.6, this will conclude the proof.

Initially, we have to check (33) for vertices p of zero in-degree and $\sigma \in S(p)$. If $p \notin X$, there is nothing to prove. If $p \in X$, we have $\mathrm{at}'(p,\sigma) = \mathrm{at}(p,\sigma) + d(p) \leq \mathrm{rat}(p,\sigma) = \mathrm{rat}(p,\sigma) - (1/\nu_p)(\mathrm{slew}'(p,\sigma) - \mathrm{slew}(p,\sigma))$. In either case, the claim holds.

Suppose now that the claim holds at some stage of iterative propagation, and we now update $\mathrm{at}(q,\sigma')$ and $\mathrm{slew}(q,\sigma')$ for some vertex q and $\sigma' \in S(q)$ as in (17). We have to show that the claim continues to hold for q and σ'. We consider three cases.

Case 1) q is not reachable from X. Then, (32) holds for each predecessor of q, and thus it continues to hold also for q.

Case 2) $q \in X$. Then, (32) holds for each predecessor and thus we get $\mathrm{at}'(q,\sigma) = \mathrm{at}(q,\sigma) + d(q) \leq \mathrm{rat}(q,\sigma) = \mathrm{rat}(q,\sigma) - (1/\nu_q)(\mathrm{slew}'(q,\sigma) - \mathrm{slew}(q,\sigma))$, i.e., (33) holds.

Case 3) $q \notin X$, but q is reachable from X. Let $(e,\sigma) \in \Xi(q,\sigma')$, $e = (p,q)$. If p is not reachable from X [and thus satisfies (32)], we have

$$\begin{aligned}&\mathrm{at}'(e,\sigma)\\ &\quad= \mathrm{at}'(p,\sigma) + \vartheta_e\,(\mathrm{slew}'(p,\sigma))\\ &\quad= \mathrm{at}(p,\sigma) + \vartheta_e\,(\mathrm{slew}(p,\sigma))\\ &\quad\leq \mathrm{at}(q,\sigma') - \frac{1}{\nu_q}\max\{0, \lambda_e\,(\mathrm{slew}(p,\sigma)) - \mathrm{slew}(q,\sigma')\}\\ &\quad\leq \mathrm{rat}(q,\sigma') - \frac{1}{\nu_q}\max\{0, \lambda_e\,(\mathrm{slew}(p,\sigma)) - \mathrm{slew}(q,\sigma')\}\\ &\quad= \mathrm{rat}(q,\sigma') - \frac{1}{\nu_q}\max\{0, \lambda_e\,(\mathrm{slew}'(p,\sigma)) - \mathrm{slew}(q,\sigma')\}\\ &\quad= \mathrm{rat}(q,\sigma') - \frac{1}{\nu_q}\max\{0, \mathrm{slew}'(e,\sigma) - \mathrm{slew}(q,\sigma')\}.\end{aligned} \tag{34}$$

If p be reachable from X [and thus satisfies (33)], we have (abbreviating $M := \max\{0, \mathrm{slew}'(p) - \mathrm{slew}(p)\}$)

$$\begin{aligned}\mathrm{at}'(e,\sigma) &= \mathrm{at}'(p,\sigma) + \vartheta_e\,(\mathrm{slew}'(p,\sigma))\\ &\leq \mathrm{rat}(p,\sigma) - \frac{1}{\nu_p}M + \vartheta_e\,(\mathrm{slew}'(p,\sigma))\\ &\leq \mathrm{rat}(e,\sigma') - \frac{1}{\nu_p}M + \vartheta_e\,(\mathrm{slew}(p,\sigma)) + \hat{\vartheta}_e M\\ &= \mathrm{rat}(q,\sigma') - \frac{1}{\nu_q}\max\{0, \lambda_e\,(\mathrm{slew}(p,\sigma)) - \mathrm{slew}(q,\sigma')\}\\ &\quad + \left(\hat{\vartheta}_e - \frac{1}{\nu_p}\right)M\\ &\leq \mathrm{rat}(q,\sigma') - \frac{1}{\nu_q}\max\{0, \lambda_e\,(\mathrm{slew}'(p,\sigma)) - \mathrm{slew}(q,\sigma')\}\\ &\quad + \frac{\hat{\lambda}_e}{\nu_q}M + \left(\hat{\vartheta}_e - \frac{1}{\nu_p}\right)M\\ &= \mathrm{rat}(q,\sigma') - \frac{1}{\nu_q}\max\{0, \mathrm{slew}'(e,\sigma) - \mathrm{slew}(q,\sigma')\}\\ &\quad + \left(\hat{\vartheta}_e + \frac{\hat{\lambda}_e}{\nu_q} - \frac{1}{\nu_p}\right)M\\ &\leq \mathrm{rat}(q,\sigma') - \frac{1}{\nu_q}\max\{0, \mathrm{slew}'(e,\sigma) - \mathrm{slew}(q,\sigma')\}\end{aligned} \tag{35}$$

as ν_x $(x \in V(G))$ are strongly safe. In either case, we conclude that $\mathrm{at}'(e,\sigma) \leq \mathrm{rat}(q,\sigma') - (1/\nu_q)\max\{0, \mathrm{slew}'(e,\sigma) - \mathrm{slew}(q,\sigma')\}$. By (17), this implies

$$\mathrm{at}'(q,\sigma') = \max\{\mathrm{at}'(e,\sigma) | (e,\sigma) \in \Xi(q,\sigma')\} \leq \mathrm{rat}(q,\sigma') \tag{36}$$

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and

$$\begin{aligned}\mathrm{at}'(q,\sigma') &= \max\left\{\mathrm{at}'(e,\sigma)+\frac{1}{\nu_q}\left(\mathrm{slew}'(e,\sigma)-\mathrm{slew}'(q,\sigma')\right)\middle|\right.\\ &\qquad\left.(e,\sigma)\in\Xi(q,\sigma')\right\}\\ &\le \max\left\{\mathrm{rat}(q,\sigma')-\frac{1}{\nu_q}\left(\mathrm{slew}'(e,\sigma)-\mathrm{slew}(q,\sigma')\right)\right.\\ &\qquad+\frac{1}{\nu_q}\left(\mathrm{slew}'(e,\sigma)-\mathrm{slew}'(q,\sigma')\right)\,|\\ &\qquad\left.(e,\sigma)\in\Xi(q,\sigma')\right\}\\ &= \mathrm{rat}(q,\sigma')-\frac{1}{\nu_q}\left(\mathrm{slew}'(q,\sigma')-\mathrm{slew}(q,\sigma')\right) \qquad (37)\end{aligned}$$

as required. ■

An analogous proof yields the corresponding statement for early slacks (we omit the details).

Theorem 3.2: Consider a timing graph where all early slacks are nonnegative. Let X be a set of vertices such that no propagation sequence contains more than one of them. Suppose we subtract a delay of $d(x) \le \mathrm{slack}(x, \mathrm{early})$ at each vertex $x \in X$. Then all early slacks with respect to the solution to the modified instance of the arrival time problem remain nonnegative. ■

Of course, these theorems are not applicable to any vertex that lies on a cyclic propagation sequence. The slack of these vertices may overestimate the delay that one can safely add at this vertex. To correct this, one could replace the late slack by the minimum of the above-defined slack and the minimum weight of cyclic late propagation sequence containing this vertex, where the weight of an arc $((p,\sigma),(q,\sigma'))$ of the late propagation graph is

$$-\vartheta_e\left(\mathrm{slew}(p,\sigma)\right)-\frac{1}{\nu_q}\max\left\{0,\lambda_e\left(\mathrm{slew}(p,\sigma)\right)-\mathrm{slew}(q,\sigma')\right\} \qquad (38)$$

where $e=(p,q)$, similar for early slacks.

D. Slack Distribution

Often, we need independent slacks on gate arcs only or on net arcs only. This means that additional delays up to such late slacks should be acceptable simultaneously at all vertices. Of course, these slacks should be "distributed as equally as possible," possibly proportionally to their sensitivity. Such a slack distribution has many applications in physical design, in particular at later stages (detailed placement, routing). See the references at the end of this section.

Consider an instance of the general static timing analysis problem, where the numbers ν_x $(x \in V(G))$ are strongly safe, and a solution to the corresponding instance of the arrival time problem. We construct the following weighted directed graph.

Definition 3.1: The balance graph consists of the vertices of the propagation graph plus an extra vertex 0 and the following arcs.

1) An arc $e'=((x',\sigma'),(x,\sigma))$ for each late arc $((x,\sigma),(x',\sigma'))$ of the propagation graph. We set $c(e') := -\vartheta_e(\mathrm{slew}(x,\sigma)) -(1/\nu_{x'})\max\{0,\lambda_e(\mathrm{slew}(x,\sigma))-\mathrm{slew}(x',\sigma')\}$, where $e=(x,x')$.
2) Each early arc $e'=((x,\sigma),(x',\sigma'))$ of the propagation graph. We set $c(e') := \vartheta_e(\mathrm{slew}(x,\sigma)) + (1/\nu_{x'})\min\{0,\lambda_e(\mathrm{slew}(x,\sigma))-\mathrm{slew}(x',\sigma')\}$, where $e=(x,x')$.

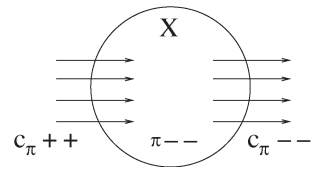

Fig. 5. Illustration of Theorem 3.3.

3) An arc $e'=((x,\sigma),0)$ for each vertex $x \in V(G)$ with zero in-degree and each late signal $\sigma \in S(x)$. We set $c(e') := -\mathrm{at}(x,\sigma)$.
4) An arc $e'=(0,(x,\sigma))$ for each vertex $x \in V(G)$ with zero in-degree and each early signal $\sigma \in S(x)$. We set $c(e') := \mathrm{at}(x,\sigma)$.
5) An arc $e'=(0,(x,\sigma))$ for each vertex $x \in V(G)$ with zero out-degree and each late signal $\sigma \in S(x)$. We set $c(e') := \mathrm{rat}(x,\sigma)$.
6) An arc $e'=((x,\sigma),0)$ for each vertex $x \in V(G)$ with zero out-degree and each early signal $\sigma \in S(x)$. We set $c(e') := -\mathrm{rat}(x,\sigma)$.
7) An arc $e'=((x,\sigma),(x',\sigma'))$ for each test. We set $c(e') := 0$.

The motivation is the following easy observation.

Proposition 3.4: Let (B,c) be the balance graph and $\pi: V(B) \to \mathbb{R}$ with $\pi(0)=0$. Then, π is a feasible potential of (B,c), i.e., $c_\pi(e) := c(e)+\pi(x)-\pi(y) \ge 0$ for $e=(x,y) \in E(B)$, if and only if after replacing arrival times by $\pi((x,\sigma))$, $x \in V(G)$, $\sigma \in S(x)$, in the solution to the arrival time problem it is (still) a feasible solution to the given instance of the general static timing analysis problem. ■

We call $c_\pi(e)$ the independent slack of $e=(x,y) \in E(B)$ with respect to π. However, several arcs of the balance graph stem from the same propagation segment, and there is no point in having different slacks on one and the same arc of G. In general, we have a partition of the arc set into bunches and ask for a common slack for each bunch.

Moreover, we often need independent slacks on a subset of the arcs only. Among these, the slacks should be "distributed as equally as possible," possibly proportional to their sensitivity. More precisely, we consider the following quite general combinatorial optimization problem.

General Balance Potential Problem

INSTANCE:

A directed graph B with arc lengths $c: E(B) \to \mathbb{R}$. A family $\mathcal{F}$ of disjoint subsets of $E(B)$. Weights $w: \bigcup \mathcal{F} \to \mathbb{R}_+ \setminus \{0\}$.

TASK:

Find numbers $\pi: V(B) \to \mathbb{R}$ with

$$c_\pi(e) \ge 0 \quad \text{for } e \in E(B) \setminus \bigcup \mathcal{F} \qquad (39)$$

such that the vector

$$\left(\min\left\{\frac{c_\pi(e)}{w(e)}\middle| e \in F\right\}\right)_{F\in\mathcal{F}} \qquad (40)$$

where the entries are to be sorted in nondecreasing order, is lexicographically maximum.

For the special case of unit weights, Albrecht *et al.* [3] have described an $O(mn+n^2\log n)$ algorithm, extending the parametric

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shortest path algorithm of Young *et al.* [19]. For the generalization to arbitrary weights, see [8].

Other works considered mostly the trivial partition of all singletons. If $w \equiv 1$, this special case reduces to the so-called minimum balance problem, which has been considered by Schneider and Schneider [16] and Young *et al.* [19].

The lexicographic order is the appropriate balance measure, as is demonstrated by the following equivalent formulation.

Theorem 3.3: Let $(B, c, \mathcal{F}, w)$ be an instance of the general balance potential problem. For $\pi : V(B) \to \mathbb{R}$, let

$$E_\pi := \left\{ f \in F \in \mathcal{F} \,\middle|\, \frac{c_\pi(f)}{w(f)} = \min \left\{ \frac{c_\pi(e)}{w(e)} \,\middle|\, e \in F \right\} \right\}. \quad (41)$$

Let $\pi : V(B) \to \mathbb{R}$ with (39). Then, π is an optimum solution for which E_π is minimal among all optimal solutions if and only if for each $X \subseteq V(B)$

$$\min_{e=(x,y)\in\delta_B^+(X)\setminus\cup\mathcal{F}} c_\pi(e) = 0$$

or

$$\min_{e\in E_\pi\cap\delta_B^-(X)} \frac{c_\pi(e)}{w(e)} \geq \min_{e\in E_\pi\cap\delta_B^+(X)} \frac{c_\pi(e)}{w(e)} \quad (42)$$

whereas usually $\min \emptyset = \infty$. Moreover, the set E_π and the vector $(c_\pi(e))_{e\in E_\pi}$ are identical for all optimum solutions π.

Note that (42) can be interpreted as a local optimality criterion: reducing π on any set X [and thus decreasing $c_\pi(e)$ for $e \in \delta_B^+(X)$ and increasing $c_\pi(e)$ for $e \in \delta_B^-(X)$] makes the situation worse (cf. Fig. 5).

Proof: Any optimal solution with minimal E_π satisfies (42); otherwise, we could decrease $\pi(x)$ $(x \in X)$ by a sufficiently small $\epsilon > 0$ and thereby increase the sorted vector (40) lexicographically (disproving optimality), or keep this vector identical (maintaining optimality) but remove at least one arc from E_π.

Let now $\pi, \pi' : V(B) \to \mathbb{R}$ be two vectors with (39) and (42), and suppose that there exists an $f = (p, q) \in E_\pi \cup E_{\pi'}$ with $\pi(p) - \pi(q) \neq \pi'(p) - \pi'(q)$, and choose f such that $\min\{c_\pi(f), c_{\pi'}(f)\}/w(f)$ is minimum. Without loss of generality, $\pi(p) - \pi(q) < \pi'(p) - \pi'(q)$. Then, $f \in E_\pi$: the contrary assumption would imply that there exist $f' \in E(B)$ and $F \in \mathcal{F}$ with $f, f' \in F$ and $c_\pi(f')/w(f') < c_\pi(f)/w(f)$. Then, $c_{\pi'}(f')/w(f') = c_\pi(f')/w(f') < c_\pi(f)/w(f) < c_{\pi'}(f)/w(f)$ and thus $f \notin E_{\pi'}$, a contradiction.

Let Q be the set of all vertices reachable from q via arcs $e \in E(B) \setminus \bigcup \mathcal{F}$ with $c_\pi(e) = 0$ or arcs $e \in E_\pi$ with $c_\pi(e)/w(e) \leq c_\pi(f)/w(f)$. Then, by (42), we have $p \in Q$. Hence, there is a q–p path P that consists of arcs $e = (x, y)$ with $c(e) + \pi(x) - \pi(y) \leq c(e) + \pi'(x) - \pi'(y)$. Summation yields $\pi(q) - \pi(p) \leq \pi'(q) - \pi'(p)$, a contradiction. ■

Special cases of this theorem have been proved by Schneider and Schneider [16] and Albrecht [2].

Other approaches to slack balancing known from the literature are the so-called zero-slack algorithm [14], the extensions by Youssef *et al.* [20], and the algorithm by Chen *et al.* [7] (see also [6]) that distributes slacks in order to maximize the sum of all independent slacks. They all work for acyclic timing graphs only. See also [12]. There are many other papers on slack balancing, or, almost equivalently, delay budgeting, with slightly different objectives, new applications, and combinations with other design tasks, in particular placement or retiming. See [5] and [18] for examples.

IV. Conclusion

Classical static timing analysis is simple and efficient but has several major drawbacks: it may conclude that a chip works although it does not, results are not continuous functions of the input (in contrast to reality), convergence in nonacyclic timing graphs cannot be proved, and slacks are often too optimistic. We proposed a modification, based on enhanced slew propagation, that is just as efficient but has none of these drawbacks. In particular, we redefined the notion of slack, which is central in very large scale integration (VLSI) design, properly. We also addressed how the popular method of distributing slack is affected. A thorough experimental study with an industrial timing engine and a combination with statistical timing is left for future work.

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